

AERO70038 - Problem set 3

Aerodynamics and Performance in forward flight

1. Find the iterative solution of the inflow equation using both the fixed point iterative method and the Newton-Raphson method for the cases below (to a 0.05% tolerance).
 - (a) Forward flight, with $C_T = 0.008$, $\alpha = 0$ and $\mu = 0.05$.
 - (b) Axial climb, with $C_T = 0.008$ and $\lambda_c = 0.1$.

$$(a) \alpha = 0 \quad \therefore \lambda = \frac{V_\infty \sin \alpha + v_i}{\Omega R} = \frac{v_i}{\Omega R} = \lambda_i$$

$$\therefore \lambda_i = \frac{C_T}{2\sqrt{\mu^2 + \lambda_i^2}} = \frac{0.004}{\sqrt{0.05^2 + \lambda_i^2}}$$

$$\text{let } (\lambda_i)_0 = 0$$

① iterative method, for $\epsilon = 0.001$

$$(\lambda_i)_1 = 0.08$$

$$(\lambda_i)_2 = 0.0424$$

$$(\lambda_i)_3 = 0.0610$$

$$(\lambda_i)_4 = 0.0507$$

$$(\lambda_i)_5 = 0.0562$$

$$(\lambda_i)_6 = 0.0532$$

$$(\lambda_i)_7 = 0.0548$$

$$(\lambda_i)_8 = 0.0539$$

$$(\lambda_i)_9 = 0.0544$$

$$\therefore \lambda = \lambda_i = 0.054$$

② Newton - Raphson method

$$f(\lambda_i) = \lambda_i - \frac{C_T}{2\sqrt{\mu^2 + \lambda_i^2}}$$

$$f'(\lambda_i) = 1 + \frac{C_T \lambda_i}{2(\mu^2 + \lambda_i^2)^{\frac{3}{2}}}$$

$$\lambda_{i+1} = \lambda_i - \frac{f(\lambda_i)}{f'(\lambda_i)}$$

$$\text{for } \epsilon = 0.0001$$

$$(\lambda_i)_1 = 0.08$$

$$(\lambda_i)_2 = 0.5278$$

$$(\lambda_i)_3 = 0.5422$$

$$(\lambda_i)_4 = 0.5423$$

$$(\lambda_i)_5 = 0.5423 \quad \therefore \lambda = \lambda_i = 0.05423$$

$$\text{b) } \alpha = 90^\circ, V_m = 0$$

$$\therefore \mu = 0 \quad \therefore \lambda = \lambda_i$$

$$\therefore \lambda_i = \lambda_c + \frac{C_T}{2\lambda_i} = 0.1 + \frac{0.004}{\lambda_i}$$

$$\text{let } (\lambda_i)_0 = 0.5$$

① iterative method, for $\epsilon = 0.001$

$$(\lambda_i)_1 = 0.108$$

$$(\lambda_i)_2 = 0.1370$$

$$(\lambda_i)_3 = 0.1292$$

$$(\lambda_i)_4 = 0.1310$$

$$(\lambda_i)_5 = 0.1305$$

$$(\lambda_i)_6 = 0.1306 \quad \therefore \lambda = \lambda_i = 0.131$$

② Newton-Raphson method

$$f(\lambda_i) = \lambda_i - \lambda_c - \frac{C_T}{2\lambda_i} = \lambda_i - 0.1 - \frac{0.004}{\lambda_i}$$

$$f'(\lambda_i) = 1 + \frac{C_T}{2\lambda_i^2} = 1 + \frac{0.004}{\lambda_i^2}$$

$$\lambda_{i+1} = \lambda_i - \frac{f(\lambda_i)}{f'(\lambda_i)}$$

$$\text{for } \epsilon = 0.0001$$

$$(\lambda_i)_1 = 0.1142$$

$$(\lambda_i)_2 = 0.1301$$

$$(\lambda_i)_3 = 0.1306$$

$$(\lambda_i)_4 = 0.1306 \quad \therefore \lambda = \lambda_i = 0.1306$$

$$\text{let } \lambda_0 = \lambda_h = \sqrt{\frac{C_T}{2}}$$

2. For helicopter operations at high speed, it is possible that a "reverse flow" region can exist at the root end of the blade in the retreating side of the rotor disk. Show that the reverse flow region on the rotor disk is a circle of diameter μ with centre located at $(r, \psi) = (\mu/2, 3\pi/2)$.

The velocity normal to blade in forward flight is :

$$U_T = (V_\infty)_T \sin \psi + \Omega y = V_\infty \cos \alpha \sin \psi + \Omega y$$

$$\therefore \frac{U_T}{\Omega R} = \mu \sin \psi + r$$

$$\therefore \text{reverse flow occurs when } U_T < 0$$

$$\therefore \text{let } \mu \sin \psi + r = 0 \quad \therefore r = -\mu \sin \psi$$

$$\therefore r^2 = x^2 + y^2 = (r \cos \psi)^2 + (r \sin \psi)^2$$

$$\therefore x = -\mu \cos \psi \sin \psi$$

$$y = -\mu \sin^2 \psi$$

$$\text{Also, } r^2 = \mu^2 \sin^2 \psi = -\mu y$$

$$\therefore x^2 + y^2 = -\mu y \quad \therefore x^2 + y^2 + \mu y = 0$$

$$\therefore x^2 + (y + \frac{1}{2}\mu)^2 = (\frac{1}{2}\mu)^2 \quad \therefore r = \frac{1}{2}\mu$$

$$\text{centre at } (x, y) = (0, -\mu)$$

$$\therefore \cos \psi \sin \psi = 0 \text{ and } \sin^2 \psi = 1 \quad \therefore \psi = -\frac{3}{2}\pi$$

$$\therefore \text{centre at } (r, \psi) = (\frac{1}{2}\mu, -\frac{3}{2}\pi)$$

3. Consider a single main rotor helicopter with the parameters tabulated below. The aircraft has a main gearbox with 2 stages of gear reduction and is flying at a speed of 120 kt (61.7 m s^{-1}).

- What is its power consumption? Indicate what additional assumptions are implicit if the problem is to be solved with no additional information.
- Write down the equation for power consumption in a form that applies to autorotation and calculate the aircraft descent rate in a stable autorotation, stating appropriate assumptions.

Aircraft mass m	4545 kg
Number of main rotor blades N_b	4
Blade chord c	0.4 m
Rotor radius R	6.6 m
Rotor normal operating tip speed ΩR	227 m s^{-1}
Parasite drag area f	2.5 m^2
Number of tail rotor blades N_{b, \tau}	4
Tail rotor blade chord c_{\tau}	0.2 m
Tail rotor radius R_{\tau}	1.2 m
Tail rotor tip speed $(\Omega R)_{\tau}$	210 m s^{-1}
Tail rotor offset l_{\tau}	7.8 m
Main rotor blade section drag coefficient	0.011 C_d
Tail rotor blade section drag coefficient	0.012 C_{d, \tau}
Ambient density	1 kg m^{-3}

(a) Assume small angle of attack for TPP, no power require for auxiliary system, no blockage effect, download factor is negligible, $K_h = 1.15$

$$\therefore K_{trans} = 1 + 2 \cdot 0.02 = 1.04$$

$$\mu = \frac{V_{\infty} \cos \alpha}{\Omega R} = \frac{61.7}{227} = 0.2718$$

① Induced power

$$K = K_h \cosh(7.5\mu^2) = 1.15 \cdot 1.157 = 1.331$$

$$\therefore C_T = \frac{T}{\rho A V_{tip}^2} = \frac{mg}{\rho A (\Omega R)^2} = 6.323 \times 10^{-3}$$

$$\therefore \lambda = \lambda_i = \frac{C_T}{2\mu} = 0.0116 \text{ since } \mu > 0.1$$

$$\therefore C_{Pi} = \frac{K T v_i}{\rho A (\Omega R)^2} = K C_T \lambda_i = 9.784 \times 10^{-5}$$

② profile power

$$C_D = \frac{N_b C}{\pi R} = 0.07717$$

$$K = 4.65 \text{ since } \mu < 0.5$$

$$\therefore C_{Po} = \frac{1}{8} C_D (1 + K\mu^2) = 1.426 \times 10^{-4}$$

③ parasitic power

$$C_{Pp} = \frac{1}{2} \left(\frac{f}{A} \right) \mu^3 = 1.834 \times 10^{-4}$$

∴ for main rotor.

$$P = \rho A V_{tip}^3 C_p = P_r + P_i + P_p = 6.785 \times 10^5 \text{ (W)}$$

④ tail rotor induced power

$$\therefore Q_{MR} = \frac{P_{MR}}{\Omega_{MR}} = T_{TR} l_{TR}$$

$$\therefore T_{TR} = \frac{P_{MR}}{\Omega_{MR} l_{TR}} = 2529 \text{ (N)}$$

$$\therefore C_{T,TR} = 0.01268$$

$$\mu_{TR} (\Omega R)_{TR} = \mu (\Omega R) \quad \therefore \mu_{TR} = \mu \frac{\Omega R}{(\Omega R)_{TR}} = 0.2938$$

$$\therefore \lambda_{i,TR} = \frac{C_{T,TR}}{2\mu_{TR}} = 0.02158$$

$$K = K_{h,TR} \cosh(7.5\mu_{TR}^2) = 1.2 \times 1.217 = 1.460$$

$$\therefore C_{P_i,TR} = K C_{T,TR} \lambda_{i,TR} = 3.995 \times 10^{-4}$$

⑤ tail rotor profile power

$$C_{TR} = \left(\frac{N_b C}{\pi R} \right)_{TR} = 0.2122$$

$$\therefore C_{P,TR} = \frac{1}{8} C_{TR} C_{d,TR} (1 + K/\mu_{TR}^2) = 4.461 \times 10^{-4}$$

∴ for tail rotor.

$$P_{TR} = P_{i,TR} + P_{o,TR} = 3.543 \times 10^4 \text{ (W)}$$

$$\therefore P_{total} = K_{trans} (P + P_{TR}) = 7.425 \times 10^5 \text{ (W)}$$

(b) for autorotation, $T \approx W$

$$\therefore P = T V_c \approx W V_c = mg V_c$$

$$\therefore V_c = -V_b = \frac{-P_{\text{total}}}{mg} = -16.65 \text{ (m/s)}$$

$$\therefore V_b = 16.65 \text{ (m/s)}$$

4. The rotors of a tilt rotor aircraft, with radius of 5.8 m, have been tested in isolation and have exhibited an induced power factor of 1.1. Over a wide range of thrust values that would cover the aircraft all-up weight, the rotors demonstrated a figure of merit of 0.8.

- (a) What shaft power is required for the aircraft to hover at sea level at its maximum weight of 200000 N if the wing download/fountain effect amounts to 7% of the weight?
- (b) If 6000 kW are available at the rotor shafts, what vertical climb rate can the aircraft achieve?

$$(a) \quad T = K_{\text{download}} W = 1.07 \times 2 \times 10^5 = 2.14 \times 10^5 \text{ (N)}$$

$$\therefore P_{\text{ideal}} = T v_h = \frac{T^{\frac{3}{2}}}{\sqrt{2\rho A}} = 6.150 \times 10^6 \text{ (W)}$$

$$\therefore P_{\text{actual}} = \frac{P_{\text{ideal}}}{F_M} = 7.688 \times 10^6 \text{ (W)}$$

(b) assume $P_h = 5435.6 \text{ (kW)}$ due to error in problem

$$\therefore \frac{P}{P_h} = 1 + \frac{\Delta P}{T v_h} \approx \frac{V_c}{2 v_h} + 1 \quad \text{by assuming } \left(\frac{V_c}{2 v_h}\right)^2 \ll 1$$

$$\therefore V_c = \frac{2\Delta P}{T} = \frac{2\Delta P}{1.07W} = \frac{2(P - P_h)}{1.07W}$$

$$\therefore V_c = 2.637 \text{ (m/s)}$$

Answers:

1. (a) $\lambda_i = 0.054229$
(b) $\lambda_i = 0.130622$
2. Proof.
3. (a) 742.5 kW
(b) 16.65 m/s
4. (a) 5435.6 kW
(b) 5.27 m/s